

AnimatedSwitcher – Flutter Widget Documentation

Overview

`AnimatedSwitcher` is a Flutter widget that **automatically animates when its child widget changes**. It is commonly used to smoothly transition between two UI states, such as **loading** → **content**, **empty** → **data**, or **icon/text changes**.

It works by comparing the **old child** and the **new child** and animating between them.

Basic Syntax

```
AnimatedSwitcher(  
  duration: Duration(milliseconds: 300),  
  child: isLoading  
    ? CircularProgressIndicator()  
    : Text("Done"),  
);
```

How It Works

- When `isLoading` changes:
- The old widget is animated **out**
- The new widget is animated **in**
- Flutter uses a **FadeTransition by default**

No `setState` animation code is required — Flutter handles it for you.

Required Properties

`duration`

Defines how long the animation will take.

```
Duration(milliseconds: 300)
```

child

The widget that will change and be animated.

⚠ Important: The child should be a **different widget instance** when changed.

Important Rule (Very Important ⚠)

Each child **must have a unique identity**.

✅ Best practice: Use `Key`

```
AnimatedSwitcher(  
  duration: Duration(milliseconds: 300),  
  child: isLoading  
    ? CircularProgressIndicator(key: ValueKey("loading"))  
    : Text("Done", key: ValueKey("done")),  
);
```

Without keys, Flutter may not animate correctly.

Common Use Cases

1 Loading to Content

```
AnimatedSwitcher(  
  duration: Duration(milliseconds: 300),  
  child: isLoading  
    ? CircularProgressIndicator()  
    : Text("Data Loaded"),  
);
```

2 Toggle Text

```
AnimatedSwitcher(  
  duration: Duration(milliseconds: 250),  
  child: isOn  
    ? Text("ON", key: ValueKey(true))
```

```
        : Text("OFF", key: ValueKey(false)),  
    );
```

3 Icon Change Animation

```
AnimatedSwitcher(  
    duration: Duration(milliseconds: 200),  
    child: isFavorite  
        ? Icon(Icons.favorite, key: ValueKey("fav"))  
        : Icon(Icons.favorite_border, key: ValueKey("notFav")),  
);
```

Custom Animation (Advanced)

You can customize the animation using `transitionBuilder`.

```
AnimatedSwitcher(  
    duration: Duration(milliseconds: 400),  
    transitionBuilder: (child, animation) {  
        return ScaleTransition(scale: animation, child: child);  
    },  
    child: Text("Hello", key: ValueKey("hello")),  
);
```

Common Transitions

- FadeTransition
- ScaleTransition
- SlideTransition
- RotationTransition

Performance Notes

✓ Lightweight and efficient

⚠ Avoid using it for: - Large lists - Highly complex widgets rebuilding frequently

When to Use AnimatedSwitcher

✓ Loading → Success UI ✓ Toggle buttons ✓ Status text changes ✓ Icon animations

✗ Not ideal for list item animations (use `AnimatedList` instead)

Summary

- `AnimatedSwitcher` animates widget changes automatically
 - Requires unique child identity (use `Key`)
 - Easy to implement
 - Clean and professional UI transitions
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 You can edit this document to add screenshots, notes, or your own examples.